

Ke Li

Personal Website: <https://kelichloe.github.io>

Research Interests

My research lies at the intersection of AI/ML, optimization, and decision making under uncertainty. I pursue two complementary streams:

- **Human-AI Collaboration in Scientific Process:** studying how machine learning and generative AI can augment the scientific process, particularly in theory building.
- **Data-Driven Optimization:** developing data-driven machine learning algorithms to support robust decision-making under uncertainty.

Education

INSEAD	France
<i>Ph.D. Candidate in Decision Sciences (AI/ML + Optimization)</i>	09/2023 - 06/2028
Advisor: Spyros Zoumpoulis	
The Chinese University of Hong Kong, Shenzhen	China
<i>B.Eng (Honors) in Computer Science and Engineering</i>	09/2018 - 06/2022
Alumni Interview Artical (EN version/ CN version)	
China National College Entrance Examination (Gaokao)	China
Ranked in the Top 0.1% (among ~1 million candidates) in Henan Province	2018

Honors & Awards

Teaching Excellence Award, INSEAD	2025
“Stars of Tomorrow” Award of Excellent Intern (Top 10%), Microsoft Research Asia [link]	2022
ExploreCSR Computing Research Award, Google Research & UTRGV [link]	2021
Undergraduate Research Awards, CUHK-Shenzhen	2020
Undergraduate Student Teaching Fellows, CUHK-Shenzhen	2020
China National Encouragement Scholarship, Ministry of Education of China	2019
Dean’s List, CUHK-Shenzhen	2019-20
Admission Fellowship (CNY 90,000/year), CUHK-Shenzhen	2018-22

Publications - [See Google Scholar](#)

1. Zhou, B., Li, K., Jiang, J., & Lu, Z. Learning from Visual Observation via Offline Pretrained State-to-Go Transformer. *Advances in Neural Information Processing Systems (NeurIPS, Top-tier AI/CS/ML conference)*, 2023.
2. Dong, J., Li, K., Li, S., & Wang, B. Combinatorial bandits under strategic manipulations. In *the Fifteenth ACM International Conference on Web Search and Data Mining (WSDM, Top-tier data mining conference)*, 2022.
3. Liu, Y., Li, K., Huang, Z., Li, B., Wang, G., & Cai, W. EduChain: a blockchain-based education data management system. In *Blockchain Technology and Application: Third CCF China Blockchain Conference*, 2021.

Working Papers

- **A Human-AI Collaborative Framework for Theory Building** - Joint work with [Philip Parker](#), [Phanish Puranam](#), [Eric Uhlmann](#), and [Spyros Zoumpoulis](#).

Presented at:

- Selected Presentation, AI Plus Management Doctoral Consortium, London UK, 2025
- Invited Session, INFORMS Annual Meeting, Atlanta USA, 2025

- **Optimizing for Decisions: A Decision-Aware Segmentation Approach** - Joint work with [Sandeep Chandukala](#), [Ernst Osinga](#), and [Spyros Zoumpoulis](#).

Industry Experiences

Beijing Academy of Artificial Intelligence

Beijing

Reinforcement Learning Researcher (Full-time)

12/2022 - 07/2023

- Designed and trained intelligent agents capable of learning directly from high-dimensional visual observations in complex 3D game environments such as Minecraft.
- Developed novel architectures combining convolutional encoders with deep reinforcement learning algorithms (e.g., PPO, DQN)
- Led extensive experiments to validate model performance, and collaborated closely with researchers from Peking University. Our research paper ([website](#)) is accepted by NeurIPS 2023.

Microsoft Research Asia

Beijing

Research Intern, Award of Excellent Intern, Innovation Engineering Group

01/2022 - 06/2022

- Investigated model compression techniques for Transformer/BERT models, such as pruning, quantization, and distillation.
- Implemented compression pipelines and reproduced results from recent research papers.
- Optimized Transformer models for deployment by reducing parameters and memory footprint.
- Collaborated with researchers to refine experimental settings and improve reproducibility.

SenseTime

Shenzhen

Reinforcement Learning Intern, OpenDI Lab

08/2021 - 12/2021

ByteDance Technology

Beijing

Machine Learning Intern, Data AI Lab

12/2020 - 05/2021

Contributions to Open-Source Community

- **Microsoft Research Asia** — Contributed to Microsoft's AutoML toolkit [NNI](#) (14,300+ GitHub stars), including transformer model-compression modules now integrated into both internal and external AutoML pipelines.
- **SenseTime (OpenDI Lab)** — Contributed 1,600+ lines of code to the [DI-engine](#) reinforcement-learning framework (3,600+ GitHub stars), improving algorithmic stability and large-scale training utilities.

Teaching Experience

INSEAD, Teaching Assistant

Winner of Teaching Excellence Award

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| • Foundations of AI for Managers, MBA Course | 2025.05 - 2025.06 |
| • Foundations of AI for Managers, MBA Course | 2025.01 - 2025.02 |
| • Probability and Statistics, PhD Course | 2024.10 - 2024.12 |

CUHK-Shenzhen, Undergraduate Teaching Assistant

MAT1010 Calculus I	Fall 2019-2020
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Academic Services

Session Chair (co-chair with Spyros Zoumpoulis), INFORMS Annual Meeting	2025
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Session Title: Estimation and Optimization in Data-Driven Decisions

Reviewer , NeurIPS (Top-tier conference in CS/ML/AI)	2024
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Patents

Personnel exemption aided decision making method and system based on historical big data. China Patent CN113673943A. [\[link\]](#)

Random Facts

I am a part-time model in Paris ([\[portfolio\]](#)).